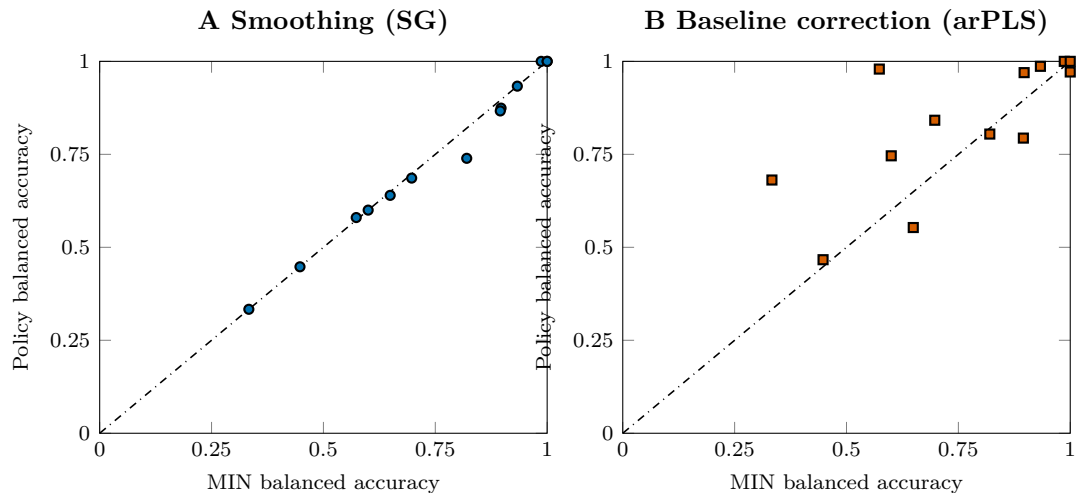




RQ-S01: preprocessing and chemical identification. Source-only selection/calibration; selected CNN identity fixed across policies. SG = impulse replacement, smoothing, minmax; arPLS = impulse replacement, baseline correction, minmax. MIN = minimal minmax. Primary: equal contexts within domain, then equal domains. Pooled-fold sensitivity reconstructs each four-fold prediction set before scoring. M01 individual-spectrum predictions; M06 combined predictions per physical sample. M06 averages model probabilities within sample/instrument then equally across instruments, never input spectra or hard labels. 598 original spectra / 69 physical samples / 10 instruments; 557 spectra in 13 held station/instrument domains, 260 contexts. Per-domain independent support in the CSV/table. D1..D13 are lexicographic station/instrument domains, not samples. Diagonal $y=x$; the 520 points across all panels are paired domain measurements, not 520 independent observations. Claim limit: cannot establish causal background removal or new-instrument performance; no error bars, no significance stars.